

Discrete-Time Fourier Series (DTFS)

- Let $x[n]$ be periodic signal N , its F.S. is

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk \frac{2\pi}{N} n} \quad \text{Synthesis Eq.}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi}{N} n} \quad \text{analysis Eq.}$$

→ spectral coefficients
F.S. coefficients

a_k is periodic
(N)

$$a_k = a_{k+N}$$

example

$$x[n] = \sin\left[\frac{\pi(n-1)}{4}\right]$$

Recall: $\sin(\Omega n)$

$$\frac{\Omega}{2\pi} = \frac{m}{N}$$

$$= \sin\left(\frac{\pi n}{4} - \frac{\pi}{4}\right)$$

$$\frac{\Omega}{2\pi} = \frac{\pi/4}{2\pi} = \frac{m}{N}$$

$$= \frac{\cancel{\pi}}{8\cancel{\pi}} = \frac{m}{N}$$

is periodic $N=8$

it can be found by $x[n+N] = x[n]$

write:

$$\begin{aligned} \sin\left(\frac{\pi(n-1)}{4}\right) &= \frac{1}{2j} e^{j \frac{\pi(n-1)}{4}} - \frac{1}{2j} e^{-j \frac{\pi(n-1)}{4}} \\ &= \frac{1}{2} e^{-j \frac{\pi}{4}} \cdot e^{j \frac{\pi}{4} n} - \frac{1}{2j} e^{+j \frac{\pi}{4}} \cdot e^{-j \frac{\pi}{4} n} \end{aligned}$$

$$e^{j \frac{2 \cdot \pi \cdot n}{2 \cdot 4}} = e^{j \frac{2\pi n}{8}}$$

Compare with $x[n] = \sum_{k=\langle N \rangle} a_k e^{jk \frac{2\pi}{N} n}$

$$= \underbrace{\frac{1}{2j} e^{-j \frac{\pi}{4}}}_{a_1} \cdot e^{j(1) \frac{2\pi}{8} n} - \underbrace{\frac{1}{2j} e^{j \frac{\pi}{4}}}_{a_{-1}} \cdot e^{j(-1) \frac{2\pi}{8} n}$$

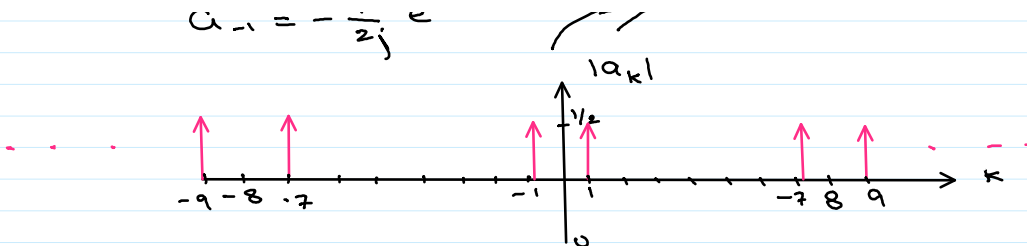
$$a_1 = \frac{1}{2j} e^{-j \frac{\pi}{4}}$$

$$a_{-1} = -\frac{1}{2j} e^{j \frac{\pi}{4}}$$

$$a_{-1} = a_1^*$$

↑ $|a_k|$

$$a_{-1} = -\frac{1}{2j} e$$



$$a_k = a_{k+8}$$

$$k=1$$

$$k=-1$$

$$a_9$$

$$a_7$$

example 2

$x[n]$ is periodic, $N=6$ and is given by

$$x[n] = \left(\frac{1}{2}\right)^n \quad -2 \leq n \leq 3$$

Find its F.S. coefficients.

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{jk \frac{2\pi}{N} n} \quad N=6$$

$$= \frac{1}{6} \sum_{n=-2}^3 \left(\frac{1}{2}\right)^n e^{jk \frac{2\pi}{6} n}$$

change :

$$m = n + 2$$

$$n = m - 2$$

$$= \frac{1}{6} \sum_{m=0}^5 \left(\frac{1}{2}\right)^{m-2} e^{-jk \frac{2\pi}{6} (m-2)}$$

$$= \frac{1}{6} \sum_{m=0}^5 \left(\frac{1}{2}\right)^{-2} e^{+jk \frac{4\pi}{6}} \left(\frac{1}{2} e^{-jk \frac{2\pi}{6}}\right)^m$$

use: $\sum_{m=0}^{N-1} \alpha^m = \frac{1 - \alpha^{N-1+1}}{1 - \alpha}$

$$= \frac{1}{6} \left(\frac{1}{4}\right)^{-1} e^{jk \frac{4\pi}{6}} \left(\frac{1 - \left(\frac{1}{2} e^{-jk \frac{2\pi}{6}}\right)^6}{1 - \frac{1}{2} e^{-jk \frac{2\pi}{6}}} \right)$$

$$= \frac{2}{3} e^{jk \frac{4\pi}{6}} \left(\frac{1 - \left(\frac{1}{2}\right)^6}{e^{+jk \frac{4\pi}{6}} \left(e^{-jk \frac{4\pi}{6}} - \frac{1}{2} e^{-jk \pi} \right)} \right)$$

$$= \frac{2}{3} \left(\frac{1 - \left(\frac{1}{2}\right)^6}{e^{-j\frac{4\pi}{6}} - \frac{1}{2}(-1)^6} \right)$$

example

$x[n]$ is a periodic signal given by

$$x[n] = (-1)^n \sin\left(\frac{6\pi n}{7}\right)$$

find a_{41} .

Solution

$$x[n] = y_1[n] \cdot y_2[n]$$

$$y_1[n] = \sin\left(\frac{6\pi n}{7}\right)$$

$$y_2[n] = (-1)^n$$

$y_1[n]$ is periodic

$$N_1 = 7$$

$$\frac{\Omega}{2\pi} = \frac{6\pi/7}{2\pi} = \frac{3}{7}$$

$y_2[n]$ is periodic

$$N_2 = 2$$

Therefore the period of $x[n]$ is

least common multiple (LCM)

$$\text{r.f.} \lceil (N_1, N_2) = 14$$

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$x[n]$ is periodic with $N = 14$

$$\omega_0 = \frac{2\pi}{N} = \frac{\pi}{7}$$

$x[n]$ is written by:

$$x[n] = \underbrace{(-1)^n}_{e^{j\pi n}} \cdot \frac{e^{j\frac{6\pi n}{7}} - e^{-j\frac{6\pi n}{7}}}{2j} = \frac{1}{2j} \left\{ e^{j\left(\pi + \frac{6\pi}{7}\right)n} - e^{j\left(\pi - \frac{6\pi}{7}\right)n} \right\}$$

$$= \frac{1}{2j} \left\{ e^{j\frac{13\pi}{14}n} - e^{j\frac{\pi}{7}n} \right\}$$

$$= \sum_{k \in \langle 14 \rangle} a_k e^{jk\omega_0 n}$$

$$\omega_0 = \frac{\pi}{7} = \frac{2\pi}{14}$$

$$\therefore a_{13} = -\frac{j}{2}, \quad a_1 = \frac{j}{2}$$

$$a_k = a_{N+k}$$

$$a_{41} = a_{41-2 \times 14} = a_{13} = -\frac{j}{2}$$

example: $x[n]$ is periodic with $N=6$

$$x[n] \xleftrightarrow{\text{F.S.}} a_k = \delta[k] - \delta[k-3] \quad 0 \leq k \leq 5$$

$y[n]$ is also periodic with $N=6$ and by;

$$y[n] = 2^n x[n] \quad 0 \leq n \leq 5$$

Let: $y[n] \xleftrightarrow{\text{F.S.}} b_k$

Find b_k for $k=0$: b_0

Solution: Fundamental freq. of $x[n]$ is $\omega_0 = \frac{2\pi}{N} = \frac{\pi}{3}$

$$\begin{aligned} x[n] &= \sum_{k=0}^{N-1} a_k e^{jk\omega_0 n} \\ &= \sum_{k=0}^5 (\delta[k] - \delta[k-3]) e^{jk \frac{\pi}{3} n} \\ &= 1 - e^{j\pi n} = 1 - (-1)^n \end{aligned}$$

so: $y[n] = 2^n x[n] = 2^n (1 - (-1)^n)$

From this eg. $y[0] = y[2] = y[4] = 0$
 $y[1] = 2, y[3] = 16, y[5] = 64$

$$b_k = \frac{1}{N} \sum_{n=\langle N \rangle} y[n] e^{-jk \frac{2\pi}{N} n}$$

set $k=0$ for $b_0 = \frac{1}{N} \sum_{n=\langle N \rangle} y[n]$

∴ $b_0 = \frac{1}{6} (y[1] + y[3] + y[5]) = 14$

example: $x[n]$ is periodic, and is given below

$$x[n] = \sum_{k=-\infty}^{+\infty} \left[(-1)^k \delta[n-k] \right] + \cos\left(\frac{2\pi}{3}n\right)$$

Find a_{21} .

Solution: use sifting property of $\delta[n]$

$$\sum_{k=-\infty}^{+\infty} (-1)^k \delta[n-k] = (-1)^n$$

∴ $x[n] = \underbrace{(-1)^n}_{N_1=2} + \underbrace{\cos\left(\frac{2\pi}{3}n\right)}_{N_2=3}$

$$x[n] = \underbrace{(-1)^n}_{N_1=2} + \underbrace{\cos\left(\frac{2\pi}{3}n\right)}_{N_2=3}$$

$$\text{r.f. } (2,3) = 6$$

$x[n]$ is periodic $N=6$

$$\omega_0 = \frac{2\pi}{N} = \frac{2\pi}{6} = \frac{\pi}{3}$$

$$x[n] = e^{j\frac{3\pi}{3}n} + \frac{1}{2}e^{j\frac{2\pi}{3}n} + \frac{1}{2}e^{j\frac{(-2)\pi}{3}n}$$

$$\Rightarrow a_k = \begin{cases} 1 & k=3 \\ \frac{1}{2} & k=\pm 2 \\ 0 & \text{o/w} \end{cases}$$

$$a_{21} = a_{21-3 \times 6} = a_3 = 1$$

example: $x_1[n]$ & $x_2[n]$ are periodic $N=4$, and shown



$$\text{Let } y[n] = \sum_{k=0}^3 x_1[k] x_2[n-k] \quad (y[n] \text{ is also periodic, } N=4)$$

$$\text{and its sample values are } y[n] = \begin{cases} +1 & n=0,3 \\ -1 & n=1,2 \end{cases}$$

Find a (on the plot).

Solution: From $y[n] = \sum_{k=0}^3 x_1[k] x_2[n-k]$, it is obvious that

$y[n]$ is periodic convolution b/w $x_1[n]$ & $x_2[n]$

$$y[n] = x_1[n] \circledast x_2[n]$$

$$\text{Let: } x_1[n] \xleftrightarrow{\text{F.S.}} a_k ; \quad x_2[n] \xleftrightarrow{\text{F.S.}} b_k ; \quad y[n] \xleftrightarrow{\text{F.S.}} c_k$$

$$\text{From convolution property } c_k = N a_k b_k$$

$$a_0 = \frac{1}{4} \sum_{n=0}^3 x_1[n] = \frac{1}{4}(1+0+2+1) = 1$$

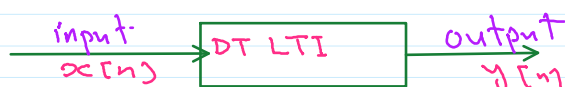
$$b_0 = \frac{1}{4} \sum_{n=0}^3 x_2[n] = \frac{1}{4}(0+1+0+a) = \frac{a+1}{4}$$

$$c_0 = \frac{1}{4} \sum_{n=0}^3 y[n] = \frac{1}{4}(y[0]+y[1]+y[2]+y[3]) = \frac{1}{4}(1-1-1+1) = 0$$

$$c_0 = 4a_0b_0 \Rightarrow c_0 = 0 = 4(1)\left(\frac{a+1}{4}\right)$$

$$a+1=0 \Rightarrow \boxed{a=-1}$$

D.T. LTI system



impulse Response
 $h[n]$

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k] h[n-k] = \sum_{k=-\infty}^{+\infty} h[k] x[n-k]$$

Let $x[n] = e^{j\Omega n}$

output :

$$y[n] = \sum_{k=-\infty}^{+\infty} h[k] e^{j\Omega(n-k)}$$

$$= \underbrace{e^{j\Omega n}}_{\text{eigenfunction}} \cdot \underbrace{\sum_{k=-\infty}^{+\infty} h[k] e^{-j\Omega k}}_{\text{eigenvalue}} = H(e^{j\Omega})$$

$$= H(e^{j\Omega}) \cdot e^{j\Omega n}$$

if $x[n] = z^n$ z is complex

then $y[n] = z^n \cdot H(z)$ where $H(z) = \sum_{k=-\infty}^{+\infty} h[k] z^{-k}$

Now if $x[n] = \sum_{k \in \langle N \rangle} a_k e^{j k \frac{2\pi}{N} n}$

due to the Linearity of this system, its output is

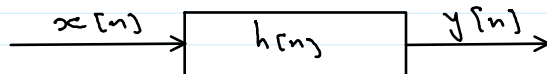
$$y[n] = \sum_{k \in \langle N \rangle} a_k H(e^{j\Omega_k}) e^{j k \frac{2\pi}{N} n}$$

where :

$$H(e^{j\Omega_k}) = \sum_{k=-\infty}^{+\infty} h[k] e^{-j\Omega_k k}$$

$$\Omega_k = k \frac{2\pi}{N}$$

example:



let : $h[n] = \alpha^n u[n]$ $|\alpha| < 1$

$x[n] = \cos\left(\frac{2\pi n}{N}\right)$ is periodic N

Find F.S. coefficients of $y[n]$.

Solution:

$$\begin{aligned} H(e^{j\Omega}) &= \sum_{n=-2}^{+2} h[n] e^{-j\Omega n} \\ &= \sum_{n=0}^2 (\alpha)^n e^{-j\Omega n} = \sum_{n=0}^{+2} (\alpha e^{-j\Omega})^n \\ &= \frac{1}{1 - \alpha e^{-j\Omega}} \quad \Omega = \frac{2\pi K}{N} \end{aligned}$$

$$x[n] = \cos\left(\frac{2\pi n}{N}\right) = \frac{1}{2} e^{j \frac{2\pi}{N} n} + \frac{1}{2} e^{-j \frac{2\pi}{N} n}$$

$$a_1 = \frac{1}{2} \quad a_{-1} = \frac{1}{2}$$

$$\therefore y[n] = \sum_{k \in \langle N \rangle} a_k H(e^{j\Omega}) \bigg|_{\Omega = \frac{2\pi k}{N}} e^{j\Omega \left(\frac{2\pi}{N}\right)n}$$

$$y[n] = \frac{1}{2} \mathcal{H} \left(e^{j \frac{2\pi}{N} n} \right) e^{j \frac{2\pi}{N} n} + \frac{1}{2} \mathcal{H} \left(e^{-j \frac{2\pi}{N} n} \right) e^{-j \frac{2\pi}{N} n}$$

$$= \underbrace{\frac{1}{2} \left(\frac{1}{1 - \alpha e^{-j 2\pi/N}} \right)}_{b_1} e^{j \frac{2\pi}{N} n} + \underbrace{\frac{1}{2} \left(\frac{1}{1 - \alpha e^{j 2\pi/N}} \right)}_{b_{-1}} e^{-j \frac{2\pi}{N} n}$$

Can be written as:

$$b_1 = r e^{j\theta}$$

Then:

$$y[n] = r \cos\left(\frac{2\pi}{N}n + \theta\right)$$

* F.S. properties:

$x[n]$ periodic N $\longleftrightarrow$ F.T. a_k

$$\text{Periodic } N \quad x[n-n_0] \xleftrightarrow{\text{F.S.}} a_k \cdot e^{-jk \frac{2\pi}{N} n}$$

$$x[-n] \xleftrightarrow{\text{F.S.}} a_{-k}$$

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$$\begin{aligned}
 & x[n] \longleftrightarrow a_k \\
 & (-1)^n x[n] \longleftrightarrow ? b_k \\
 & b_k = \frac{1}{N} \sum_{n=\langle N \rangle} \left(e^{-j\pi n} \right) x[n] \cdot e^{-jk \frac{2\pi}{N} n} \\
 & \quad \quad \quad \uparrow \\
 & \quad \quad \quad e^{-j \frac{2\pi}{N} \left(k + \frac{N}{2} \right) n} \\
 & = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] \cdot e^{-j \frac{2\pi}{N} \left(k + \frac{N}{2} \right) n} \\
 & = a_{k + \frac{N}{2}}
 \end{aligned}$$

(ex) $x[n]$ is periodic (N) $\longleftrightarrow a_k$

Let:

$$y[n] = \begin{cases} x[n] & n = \text{even} \\ 0 & n = \text{odd} \end{cases}$$

$$= \frac{1}{2} \left\{ x[n] + (-1)^n x[n] \right\} \longleftrightarrow b_k ?$$

is $y[n]$ periodic? Find its F.S. coefficients.

Sol. if N even then $y[n]$ is periodic (N)

if N odd " " " " ($2N$)

+ for N even

$$b_k = \frac{1}{2} \left\{ a_k + a_{\left(k - \frac{N}{2}\right)} \right\}$$

+ for N odd

$$b_k = \frac{1}{2N} \sum_{n=\langle 2N \rangle} y[n] e^{-jk \frac{2\pi}{2N} n}$$

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